

Shane Faberman

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[LinkedIn](#) | [Github](#) | [Personal Website/Portfolio](#)

Education

University of North Carolina | Chapel Hill, NC

August 2023 – May 2027

Bachelor of Science in Data Science and Bachelor of Science in Statistics & Analytics | GPA: 3.98/4.00

Skills

R, R Shiny, Python, SQL, Excel, Adobe Customer Journey Analytics, Google Analytics 4, Looker, Tableau, Julia, Qlik Sense

Related Experience

NBA | Basketball Strategy and Growth Intern

June 2026 – August 2026

- Developed machine learning models using **R** to estimate player substitution probabilities across millions of player-possession observations using large-scale internal NBA data.
- Architected anomaly detection frameworks to identify unusual coaching decisions relative to coach-specific historical patterns.
- Built the NBA's first ever way to monitor coaching integrity: a **React** application that updates daily in-season; operationalizing coaching anomaly models, surfacing flagged rotation decisions, and alerting the Basketball Integrity team to potential concerns via a Slack bot.

TELUS Digital | Analytics Intern

May 2025 – August 2025

- Designed, implemented, and tested a unified, scalable schema for a global hospitality client's web and mobile applications using **Adobe tools**.
- Identified over 1.2 million misclassified web analytics events and over 15,000 rows of inaccurate revenue data in a client database, and developed a bot classification model to improve their bot recognition using **SQL**.
- Improved a company-wide **Looker** dashboard used by survey captains to analyze biweekly team sentiment and inform workflow improvements across all projects.

UNC Sports Analytics Intelligence Lab | Undergraduate Student Analyst

September 2024 – Present

- Diagnose and investigate ways to improve athlete training and team performance across UNC Division 1 athletics.
- Clearly and simply articulate technical sports analytics research to coaches without analytical expertise.
- Construct informative and interactive visualizations and mathematical models, and analyze complex datasets using software such as **R**.

UNC School of School of Data and Information Sciences | Undergraduate Learning Assistant

August 2026 – Present

- Hold weekly office hours supporting 50+ students in Introduction to Python programming, data wrangling, statistical analysis, and visualization
- Grade programming assignments, quizzes, and final projects, providing detailed feedback on code quality and reasoning

QlikTech Inc. — Analytics Intern

March 2022 – March 2024 (Seasonal)

- Contributed to the creation and editing of technical documentation for annual software product conference for 3000 users.
- Co-created an interactive customer relationship analysis application in **QlikTech**, leveraging business intelligence techniques to monitor key metrics such as website session traffic.
- Used **Excel** to organize thousands of employee equity documents to ensure the accuracy and completeness of each employee's files.

Projects

NBA Coaching Impact Research

- Collected player and coaching data by web scraping; cleaned large datasets using **R** (dplyr, tidyr, and other packages).
- Modeled Expected BPM (EBPM) using inputs such as player aging trends to create a new metric (BPM Over Expected), grouped by coaches and created informative visualizations to identify insights in coaching impact.
- After presenting the research at UNC's Undergraduate Research Celebration, it was selected to be profiled on [UNC's website](#), and accepted to present at the [New England Symposium on Statistics in Sports](#) in September 2025.

College Basketball RAPM

- Architected an end-to-end **Python** pipeline to scrape and process 10GB+ of raw NCAA play-by-play data.
- Applied machine learning and Bayesian statistical techniques to compute a player's Regularized Adjusted Plus-Minus (RAPM), an estimate of impact on the success of a player's team.
- Engineered a multi-year interactive dashboard using **R Shiny**, allowing for real-time filtering and visualization of RAPM across 360+ men's and women's Division I teams.

2024 NFLPA Analytics Case Competition

- Conducted quintile-based salary analysis using **R**; found a 9.2% decline in middle-class players' revenue share from 2010 to 2024.
- Assessed labor market effects of 2020 CBA, identifying causal links between policy changes and the immediate increase in the proportion of veteran players in the NFL.
- Developed recommendations for future CBAs, including NBA style-max contract provisions and increased performance-based pay incentives.

NBA Draft Lottery Simulator

- Developed an interactive app using **R Shiny**, enabling users to visualize team pick probabilities in custom scenarios using up-to-date web-scraped lottery odds.
- Implemented user-controlled constraints to force specific teams into picks, recalculating and updating lottery odds in real time.
- Leveraged data visualization techniques to create an engaging tool for fans to explore draft scenarios and understand lottery dynamics.